Mellin transforms for multiple Jacobi-Piñeiro polynomials

Christophe Smet and Walter Van Assche
Katholieke Universiteit Leuven

August 6, 2008

Abstract

This work treats the Mellin transform of multiple Jacobi-Piñeiro polynomials. This allows us to put a number of irrationality and $\mathbb{Q}$-linear independence proofs into the framework of Hermite-Padé approximation. A similar approach is presented for the q-analogue: the q-Mellin transform of multiple little q-Jacobi polynomials and its applications in irrationality proofs.

1 Introduction

During the past decades an interesting extension of orthogonal polynomials received renewed attention. Multiple orthogonal polynomials are polynomials of one variable which satisfy orthogonality conditions with respect to several measures. Let $\mu_1, \ldots, \mu_r$ be r positive measures on the real line for which all the moments are finite. Let $\vec{n} = (n_1, n_2, \ldots, n_r)$ be a multi-index with $n_j \in \mathbb{N}$ for $1 \leq j \leq r$, with length $|\vec{n}| = n_1 + n_2 + \cdots + n_r$.

Definition 1.1. *The type I multiple orthogonal polynomials of multi-index $\vec{n}$ for the measure $(\mu_1, \ldots, \mu_r)$ are given by the vector $(A_{\vec{n},1}, A_{\vec{n},2}, \ldots, A_{\vec{n},r})$, where $A_{\vec{n},j}$ is a polynomial of degree at most $n_j - 1$, for which the following orthogonality conditions hold*

$$\int x^k \left(\sum_{j=1}^{r} A_{\vec{n},j}(x)\, d\mu_j(x) \right) = 0, \qquad k = 0, 1, \ldots, |\vec{n}| - 2, \tag{1.1}$$

$$\int x^{|\vec{n}|-1} \left(\sum_{j=1}^{r} A_{\vec{n},j}(x)\, d\mu_j(x) \right) = 1. \tag{1.2}$$

If the linear system of equations for the unknown coefficients of the polynomials $A_{\vec{n},j}$ has a solution, then it is unique (because of the normalization (1.2)) and in that case we call the multi-index $\vec{n}$ a normal index.

Definition 1.2. *The type II multiple orthogonal polynomial $P_{\vec{n}}$ is the monic polynomial*

of degree $|\vec{n}|$ for which the following orthogonality conditions hold:

$$\int P_{\vec{n}}(x)x^k\,d\mu_1(x) = 0, \qquad k=0,1,\ldots,n_1-1, \tag{1.3}$$

$$\vdots \tag{1.4}$$

$$\int P_{\vec{n}}(x)x^k\,d\mu_r(x) = 0, \qquad k=0,1,\ldots,n_r-1. \tag{1.5}$$

The matrix of the linear system of equations for the unknown coefficients of $P_{\vec{n}}$ is the transpose of the matrix for type I multiple orthogonal polynomials, hence if $\vec{n}$ is normal then the monic polynomial $P_{\vec{n}}$ exists and is unique. Type I and type II multiple orthogonal polynomials arise in a natural way in Hermite-Padé rational approximation (near infinity) of the r functions

$$\int \frac{d\mu_j(x)}{z-x}, \qquad z\notin\mathbb{R},\ j=1,2,\ldots,r,$$

(Nikishin and Sorokin [12]) and as such they go back to the end of the nineteenth century. They recently received renewed attention because analytic tools have become available to analyze them in more detail. In particular the asymptotic distribution of their zeros can be obtained using an equilibrium problem from logarithmitic potential theory for vectors of measures [6] [7] [12], and the asymptotic behaviour can be obtained by using a Riemann-Hilbert problem that characterizes the multiple orthogonal polynomials [18].

In this paper we will study certain multiple Jacobi and little q-Jacobi polynomials in somewhat more detail. For the multiple Jacobi polynomials, the r measures will be supported on $[0,1]$ and they are all beta distributions, but with one parameter fixed:

$$d\mu_j(x) = x^{\alpha_j}(1-x)^{\beta}dx, \qquad x\in[0,1], \tag{1.6}$$

where $\alpha_j,\beta>-1$. If we assume that $\alpha_j-\alpha_i\notin\mathbb{Z}$ whenever $i\neq j$, then all multi-indices are normal, so that all type I and type II multiple orthogonal polynomials exist [2] [17]. These polynomials were first considered by Piñeiro [13] and they are nowadays called Jacobi-Piñeiro polynomials. In Section 2 we will show that the Mellin transform of Jacobi-Piñeiro polynomials for $\beta=0$ are rational functions. The poles and zeros of the Mellin transform contain information about the parameters α_j and the type of the multiple orthogonal polynomials. In Section 3 we will apply this in a proof that $\zeta(3)$ is irrational (Apéry [1], Beukers [4]). In Section 4 we will show that a more recent proof of Ball and Rivoal [3] uses certain Jacobi-Piñeiro polynomials, and in Section 5 we indicate that the Mellin transform may be used to simplify Rivoal's proof [14] that at least one of the nine numbers $\zeta(5),\zeta(7),\zeta(9),\zeta(11),\zeta(13),\zeta(15),\zeta(17),\zeta(19),\zeta(21)$ is irrational.

In Section 6–9 we will deal with a q-analogue of these polynomials: the multiple little q-Jacobi polynomials are orthogonal with respect to r measures on the q-exponential lattice $\{q^k|k\in\mathbb{N}_0\}$ on $[0,1]$ and these measures have the form

$$d\mu_j(x) = x^{\alpha_j}\frac{(qx;q)_\infty}{(q^{\beta+1}x;q)_\infty}d_qx, \qquad x\in[0,1], \tag{1.7}$$

where $\alpha_j,\beta>-1$ and $(a;q)_n=\prod_{j=0}^{n-1}(1-aq^j)$. If we assume that $\alpha_j-\alpha_i\notin\mathbb{Z}$ whenever $i\neq j$, then all multi-indices are again normal. Section 6 will be the q-version of Section 2, with the Mellin transforms of type I and II multiple little q-Jacobi polynomials. In other sections we show how rational approximations to q-series such as $\zeta_q(s)$ can be seen in the framework of Mellin transforms of multiple little q-Jacobi polynomials [9].

2 Mellin transforms for Jacobi-Piñeiro polynomials

We will now consider Jacobi-Piñeiro polynomials with $\beta = 0$, i.e., we consider weights on $[0,1]$ with density $w_j(x) = x^{\alpha_j}$, $j = 1, 2, \ldots, r$, where $\alpha_j - \alpha_i \notin \mathbb{Z}$. Their Mellin transforms turn out to be rational functions. The Mellin transform $\hat{f}(s)$ of a measurable function f on $[0,\infty)$ is given by

$$\hat{f}(s) = \int_0^\infty f(x) x^s \, dx,$$

and this is defined for $0 \leq \Re s \leq M$ whenever $\int_0^\delta |f(x)| \, dx < \infty$ and $\int_\delta^\infty |f(x)| x^M \, dx < \infty$ for every $\delta > 0$. In this paper we always use functions on $[0,1]$ so that Mellin transforms are defined for $\Re s \geq 0$.

Theorem 2.1. *Let $(A_{\vec{n},1}, \ldots, A_{\vec{n},r})$ be the type I multiple orthogonal polynomials for Jacobi-Piñeiro weights with parameters $(\alpha_1, \ldots, \alpha_r; \beta = 0)$. Then*

$$\int_0^1 \left(\sum_{j=1}^r A_{\vec{n},j}(x) x^{\alpha_j} \right) x^s \, dx = C_{\vec{n}}^I \, \frac{(-s)_{|\vec{n}|-1}}{\prod_{j=1}^r (s+\alpha_j+1)_{n_j}}, \qquad \Re s \geq 0, \tag{2.1}$$

where $(a)_n = a(a+1)(a+2)\cdots(a+n-1)$ are the Pochhammer symbols (rising factorials) and

$$C_{\vec{n}}^I = (-1)^{|\vec{n}|-1} \frac{\prod_{j=1}^r (|\vec{n}| + \alpha_j)_{n_j}}{(|\vec{n}|-1)!}. \tag{2.2}$$

Proof. Let us denote the polynomial $A_{\vec{n},j}$ by

$$A_{\vec{n},j}(x) = \sum_{\ell=0}^{n_j-1} a_\ell(\vec{n}, j) x^\ell,$$

then for $\Re s \geq 0$

$$\int_0^1 A_{\vec{n},j}(x) x^{\alpha_j} x^s \, dx = \sum_{\ell=0}^{n_j-1} \frac{a_\ell(\vec{n}, j)}{s + \ell + \alpha_j + 1} = \frac{\pi_{n_j-1}(s; \vec{n}, j)}{(s+\alpha_j+1)_{n_j}},$$

where π_{n_j-1} is some polynomial of degree $n_j - 1$. This means that

$$\begin{aligned} \int_0^1 \left(\sum_{j=1}^r A_{\vec{n},j}(x) x^{\alpha_j} \right) x^s \, dx &= \sum_{j=1}^r \frac{\pi_{n_j-1}(s; \vec{n}, j)}{(s+\alpha_j+1)_{n_j}} \\ &= \frac{\pi_{|\vec{n}|-1}(s)}{(s+\alpha_1+1)_{n_1}(s+\alpha_2+1)_{n_2} \cdots (s+\alpha_r+1)_{n_r}}, \end{aligned}$$

where $\pi_{|\vec{n}|-1}$ is some polynomial of degree $|\vec{n}| - 1$. The orthogonality conditions (1.1) show that this Mellin transform is zero for $s = 0, 1, \ldots, |\vec{n}| - 2$, which gives $|\vec{n}| - 1$ zeros of the polynomial $\pi_{|\vec{n}|-1}$, and hence $\pi_{|\vec{n}|-1}(s) = Cs(s-1)(s-2)\cdots(s-|\vec{n}|+2) = (-1)^{|\vec{n}|-1}(-s)_{|\vec{n}|-1}$, with C a constant factor. The normalizing condition (1.2) shows that the Mellin transform at $s = |\vec{n}| - 1$ is equal to 1, from which the value of the constant C can be determined. □

Observe that the Mellin transform of this combination of type I Jacobi-Piñeiro polynomials is a rational function for which

- The poles are all simple and form r arithmetic sequences $\{-(\alpha_j + k) : \ k = 1, 2, \ldots, n_j\}$, $(1 \leq j \leq r)$. Hence once the poles are known, one can determine the parameters $\alpha_1, \ldots, \alpha_r$ and the multi-index $\vec{n} = (n_1, \ldots, n_r)$.
- The zeros are at the integers $0, 1, 2, \ldots, |\vec{n}| - 2$ and they give the appropriate orthogonality conditions.

The Mellin transform of the type II Jacobi-Piñeiro polynomials is given by the following theorem

Theorem 2.2. *Let $P_{\vec{n}}$ be the type II multiple orthogonal polynomial for the Jacobi-Piñeiro weights with parameters $(\alpha_1, \ldots, \alpha_r; \beta = 0)$. Then*

$$\int_0^1 P_{\vec{n}}(x) x^s \, dx = C_{\vec{n}}^{II} \ \frac{\prod_{j=1}^r (-s + \alpha_j)_{n_j}}{(s+1)_{|\vec{n}|+1}}, \qquad \Re s > 0, \tag{2.3}$$

with

$$C_{\vec{n}}^{II} = (-1)^{|\vec{n}|} \frac{|\vec{n}|!}{\prod_{j=1}^r (|\vec{n}| + \alpha_j + 1)_{n_j}}. \tag{2.4}$$

Proof. Let $P_{\vec{n}}(x) = \sum_{k=0}^{|\vec{n}|} b_k(\vec{n}) x^k$, where $b_{|\vec{n}|}(\vec{n}) = 1$, then

$$\int_0^1 P_{\vec{n}}(x) x^s \, dx = \sum_{k=0}^{|\vec{n}|} \frac{b_k(\vec{n})}{s+k+1} = \frac{\pi_{|\vec{n}|}(s)}{(s+1)_{|\vec{n}|+1}},$$

where $\pi_{|\vec{n}|}$ is some polynomial of degree $|\vec{n}|$. The orthogonality conditions (1.4) show that this Mellin transform is zero for $s = \alpha_j + k$, $k = 0, 1, \ldots, n_j - 1$ and $1 \leq j \leq r$, which gives $|\vec{n}|$ zeros for $\pi_{|\vec{n}|}$ and hence $\pi_{|\vec{n}|}(s) = C \prod_{j=1}^r (s - \alpha_j)(s - \alpha_j - 1) \cdots (s - \alpha_j - n_j + 1) = (-1)^{|\vec{n}|} \prod_{j=1}^r (-s + \alpha_j)_{n_j}$. We have normalized $P_{\vec{n}}$ to be a monic polynomial, which makes $b_{|\vec{n}|}(\vec{n}) = 1$, and since this is the residue at $s = -|\vec{n}| - 1$, we can determine the constant factor C by

$$1 = \lim_{s \to -|\vec{n}|-1} (s + |\vec{n}| + 1) \frac{\pi_{|\vec{n}|}(s)}{(s+1)_{|\vec{n}|+1}},$$

which gives the constant in (2.4). □

Observe that the Mellin transform of type II Jacobi-Piñeiro polynomials is also a rational function:

- The poles are at the negative integers $-1, -2, -3, \ldots, -|\vec{n}| - 1$, from which we can determine the length $|\vec{n}|$ of the multi-index.
- The zeros are simple and form r arithmetic sequences $\{\alpha_j + k : \ k = 0, \ldots, n_j - 1\}$ for $1 \leq j \leq r$. This allows us to determine the parameters $\alpha_1, \ldots, \alpha_r$ and the multi-index $\vec{n} = (n_1, \ldots, n_r)$. These zeros give the appropriate orthogonality conditions.

The roles of the zeros and poles are interchanged for type I and type II polynomials.

2.1 Multiple zeros or poles

If $\alpha_i - \alpha_j \notin \mathbb{Z}$ then all the poles (for type I) and the zeros (for type II) of the Mellin transform of Jacobi-Piñeiro polynomials are simple. Multiple zeros or poles appear in the confluent case when $\alpha_i - \alpha_j \to 0$. The weight functions x^{α_i} and x^{α_j} on $[0,1]$ then have to be replaced by x^{α_i} and $-\log(x)x^{\alpha_i}$ in order to have enough independent orthogonality conditions. The following result holds

Theorem 2.3. *Let $(A_{\vec{n},1}, \ldots, A_{\vec{n},r})$ be the type I multiple orthogonal polynomials for the weights*

$$\left(x^\alpha, -\log x \; x^\alpha, \ldots, (-1)^{r-1} \frac{\log^{r-1} x}{(r-1)!} \; x^\alpha \right), \qquad x \in [0,1],$$

where $\alpha > -1$. Then for $n_1 \geq n_2 \geq \cdots \geq n_r$

$$\int_0^1 \left(\sum_{j=1}^r A_{\vec{n},j}(x)(-1)^{j-1} \frac{\log^{j-1} x}{(j-1)!} \; x^\alpha \right) x^s \, dx$$
$$= C^I_{\vec{n}} \frac{(-s)_{|\vec{n}|-1}}{(s+\alpha+1)^r_{n_r} (s+\alpha+n_r+1)^{r-1}_{n_{r-1}-n_r} \cdots (s+\alpha+n_2+1)_{n_1-n_2}}. \quad (2.5)$$

Proof. Consider the Mellin transform

$$m_k(s) = (-1)^k \int_0^1 x^s \frac{\log^k x}{k!} \, dx,$$

then integration by parts easily gives $m_k(s) = m_{k-1}(s)/(s+1)$, and since $m_0(s) = 1/(s+1)$, this gives

$$m_k(s) = \frac{1}{(s+1)^{k+1}}, \qquad \Re s > -1.$$

Hence, for $A_{\vec{n},j}(x) = \sum_{\ell=0}^{n_j-1} a_\ell(\vec{n}, j) x^\ell$ we then find

$$\int_0^1 (-1)^{j-1} A_{\vec{n},j}(x) x^\alpha \frac{\log^{j-1} x}{(j-1)!} x^s \, dx = \sum_{\ell=0}^{n_j-1} \frac{a_\ell(\vec{n}, j)}{(s+\alpha+\ell+1)^j} = \frac{\pi_{j(n_j-1)}(s)}{(s+\alpha+1)^j_{n_j}},$$

where $\pi_{j(n_j-1)}$ is a polynomial of degree $j(n_j-1)$. Adding all such terms together gives

$$\int_0^1 \left(\sum_{j=1}^r A_{\vec{n},j}(x)(-1)^{j-1} \frac{\log^{j-1} x}{(j-1)!} \; x^\alpha \right) x^s \, dx$$
$$= \frac{\pi_{|\vec{n}|-1}(s)}{(s+\alpha+1)^r_{n_r} (s+\alpha+n_r+1)^{r-1}_{n_{r-1}-n_r} \cdots (s+\alpha+n_2+1)_{n_1-n_2}},$$

where $\pi_{|\vec{n}|-1}$ is a polynomial of degree $|\vec{n}|-1$. The orthogonality conditions then show that this Mellin transform is zero for $s = 0, 1, \ldots, |\vec{n}|-2$, which gives $|\vec{n}|-1$ zeros for $\pi_{|\vec{n}|-1}$. The normalization determines the constant factor. □

The condition $n_1 \geq n_2 \geq \cdots \geq n_r$ is needed in order to make

$$1, x, \ldots, x^{n_1-1}, -\log x, (-\log x)x, \ldots, (-\log x)x^{n_2-1}, \ldots, \\ (-1)^{r-1}\frac{\log^{r-1} x}{(r-1)!}, \ldots, (-1)^{r-1}\frac{\log^{r-1} x}{(r-1)!}x^{n_r-1}$$

a Chebyshev system on $[0,1]$, so that there are enough independent orthogonality conditions for the multiple orthogonal polynomials.

Theorem 2.4. *Let $P_{\vec{n}}$ be the type II multiple orthogonal polynomial for the weights*

$$\left(x^\alpha, -\log x\ x^\alpha, \ldots, (-1)^{r-1}\frac{\log^{r-1} x}{(r-1)!}\ x^\alpha\right), \qquad x \in [0,1],$$

where $\alpha > -1$. Then for $n_1 \geq n_2 \geq \cdots \geq n_r$

$$\int_0^1 P_{\vec{n}}(x)x^s\,dx = C_{\vec{n}}^{II}\frac{(-s+\alpha)_{n_r}^r(-s+\alpha+n_r)_{n_{r-1}-n_r}^{r-1}\cdots(-s+\alpha+n_2)_{n_1-n_2}}{(s+1)_{|\vec{n}|+1}}. \tag{2.6}$$

Proof. Let $P_{\vec{n}}(x) = \sum_{k=0}^{|\vec{n}|} b_k(\vec{n})x^k$, where $b_{|\vec{n}|}(\vec{n}) = 1$, then

$$\int_0^1 P_{\vec{n}}(x)x^s\,dx = \sum_{k=0}^{|\vec{n}|}\frac{b_k(\vec{n})}{s+k+1} = \frac{\pi_{|\vec{n}|}(s)}{(s+1)_{|\vec{n}|+1}},$$

where $\pi_{|\vec{n}|}$ is some polynomial of degree $|\vec{n}|$. If we take the kth derivative, then

$$\int_0^1 P_{\vec{n}}(x)x^s\left(\log^k x\right)\,dx = \frac{d^k}{ds^k}\left(\frac{\pi_{|\vec{n}|}(s)}{(s+1)_{|\vec{n}|+1}}\right).$$

The orthogonality conditions then show that $\pi_{|\vec{n}|}$ has zeros at $\alpha, \alpha+1, \ldots, \alpha+n_1-1$, the first n_2 of these zeros have at least multiplicity two, the first n_3 of these zeros have at least multiplicity three, etc. This gives all the zeros of $\pi_{|\vec{n}|}$ with their appropriate multiplicity. The normalizing condition gives the correct value of the multiplicative factor. □

2.2 Rodrigues formula

The advantage of working with the Mellin transforms of Jacobi-Piñeiro polynomials is that they are quite simple and contain the parameters and the multi-index in a very explicit way. One can derive several properties of Jacobi-Piñeiro polynomials from these Mellin transforms by using properties of the Mellin transform. The Mellin convolution $f * g$ of two functions f and g on $[0,\infty)$ is defined as

$$(f*g)(x) = \int_x^\infty f(t)g(x/t)\,\frac{dt}{t}, \qquad x \in [0,\infty),$$

and it has the property that the Mellin transform of $f * g$ is the product of the Mellin transforms $\hat{f}$ and $\hat{g}$. For functions f and g on $[0,1]$ this Mellin convolution becomes

$$(f*g)(x) = \int_x^1 f(t)g(x/t)\,\frac{dt}{t}.$$

Hence if we can recognize a Mellin transform as a product of two other Mellin transforms, then the original function is a Mellin convolution. Take for instance the Mellin transform in (2.1): this is clearly the product of r functions multiplied with an extra polynomial of degree $r-1$:

$$\frac{(-s)_{|\vec{n}|-1}}{\prod_{j=1}^{r}(s+\alpha_j+1)_{n_j}} = \left(\prod_{j=1}^{r}\frac{(-s+\sum_{i=1}^{j-1}n_i)_{n_j-1}}{(s+\alpha_j+1)_{n_j}}\right)\left(\prod_{j=1}^{r-1}(-s-1+\sum_{i=1}^{j}n_i)\right).$$

Each factor can be recognized as the Mellin transform of a Jacobi polynomial:

$$\frac{(-s+\sum_{i=1}^{j-1}n_i)_{n_j-1}}{(s+\alpha_j+1)_{n_j}} = \int_0^1 P_{n_j}^{(\alpha_j+\sum_{i=1}^{j-1}n_i,0)}(x)x^{\alpha_j}x^s\,dx,$$

where $P_n^{(\alpha,\beta)}$ is the monic Jacobi polynomial on $[0,1]$ with weight $x^\alpha(1-x)^\beta$.

Lemma 2.5. *Suppose that $f \in C^k[0,1]$ and*

$$\hat{f}(s) = \int_0^1 f(x)x^s\,dx.$$

If $f(1) = f'(1) = \ldots = f^{(k-2)}(1) = 0$ then

$$\lim_{s\to\infty} s^k\hat{f}(s) = \begin{cases} 0 & \text{if } k=0,\\ (-1)^{k-1}f^{(k-1)}(1) & \text{if } k>0.\end{cases}$$

Proof. For $k=0$ we use Lebesgue's dominated convergence theorem to find

$$\lim_{s\to\infty}\int_0^1 f(x)x^s\,dx = \int_0^1 f(x)\lim_{s\to\infty}x^s\,dx = 0.$$

For $k>0$ we use integration by parts to find

$$\int_0^1 f^{(k)}(x)x^s\,dx = f^{(k-1)}(1) + (-s)_k\int_0^1 f(x)x^{s-k}\,dx$$

from which the result follows. □

Lemma 2.6. *Suppose $f \in C^1[0,1]$ and $f(1)=0$, then*

$$(s-\alpha)\hat{f}(s) = -\widehat{D_\alpha f}(s),$$

where $(D_\alpha f)(x) = x^{-\alpha}\frac{d}{dx}x^{\alpha+1}f(x)$.

Proof. Integration by parts gives

$$\int_0^1 x^{-\alpha}\left(\frac{d}{dx}[x^{\alpha+1}f(x)]\right)x^s\,dx = -(s-\alpha)\int_0^1 x^{\alpha+1}f(x)x^{-\alpha+s-1}\,dx$$

which is the desired result. □

Lemma 2.7. *The Mellin transform of the density $(1-x)^n$ on $[0,1]$ is given by*

$$\int_0^1 (1-x)^n x^s\,dx = \frac{n!}{(s+1)_{n+1}}, \qquad \Re s > -1.$$

Proof. This Mellin transform is in fact a beta integral

$$\int_0^1 (1-x)^n x^s\,dx = \mathrm{B}(n+1,s+1) = \frac{\Gamma(n+1)\Gamma(s+1)}{\Gamma(n+s+2)},$$

and if we use $\Gamma(n+s+2) = (n+s+1)(n+s)\cdots(s+1)\Gamma(s+1)$, then the result follows. □

Theorem 2.8. *The type II Jacobi-Piñeiro polynomials are given by the following Rodrigues formula*

$$P_{\vec{n}}(x) = \frac{(-1)^{|\vec{n}|}C^{II}_{\vec{n}}}{|\vec{n}|!}\prod_{j=1}^{r}\left(x^{-\alpha_j}\frac{d^{n_j}}{dx^{n_j}}x^{\alpha_j+n_j}\right)(1-x)^{|\vec{n}|},$$

where the product of the r differential operators can be taken in any order.

Proof. The Mellin transform in (2.3) can be written as

$$\widehat{P_{\vec{n}}}(s)C^{II}_{\vec{n}}/|\vec{n}|!\ \prod_{j=1}^{r}(-s+\alpha_j)_{n_j}\ \widehat{f_{|\vec{n}|}}(s),$$

and hence by Lemmas 2.6 and 2.7 we find that $(-s+\alpha_r)_{n_r}\widehat{f_{|\vec{n}|}}(s)$ is the Mellin transform of

$$(-1)^{n_r}D_{\alpha_r}D_{\alpha_r+1}\cdots D_{\alpha_r+n_r-1}(1-x)^{|\vec{n}|}.$$

The product of the n_r differential operators simplifies to

$$D_{\alpha_r}D_{\alpha_r+1}\cdots D_{\alpha_r+n_r-1} = x^{-\alpha_r}\frac{d^{n_r}}{dx^{n_r}}x^{\alpha_r+n_r}.$$

If we repeat this procedure, then the Rodrigues formula follows. The product of the differential operators can be taken in any order since the product of the polynomial factors in the Mellin transform is commutative. □

Not only do we have this Rodrigues formula which gives an explicit expression for type I Jacobi-Piñeiro polynomials, we can also express type I and II Jacobi-Piñeiro polynomials as a convolution of the (simple) orthogonal Jacobi polynomials. In what follows, $P_n^{\alpha}(x)$ denotes the monic Jacobi polynomial with $\beta = 0$. Since a Jacobi polynomial can be seen as a special case ($r = 1$) of both type I and II Jacobi-Piñeiro polynomials, we choose its normalization as in the type II polynomials.

Theorem 2.9. *The type I Jacobi-Piñeiro polynomials are given by the following formula*

$$\sum_{j=1}^{r}A_{\vec{n},j}(x)x^{\alpha_j} = K^I_{\vec{n}}x^{-|\vec{n}|+r}\frac{d^{r-1}}{dx^{r-1}}x^{|\vec{n}|-1}$$
$$\left(P^{(\alpha_1)}_{n_1-1}(x)x^{\alpha_1} * P^{(\alpha_2+n_1-1)}_{n_2-1}(x)x^{\alpha_2} * \cdots * P^{(\alpha_r+n_1+n_2+\cdots+n_{r-1}-r+1)}_{n_r-1}x^{\alpha_r}\right)$$

with

$$K^I_{\vec{n}} = \frac{(-1)^{r-1}}{(|\vec{n}|-1)!}\prod_{j=1}^{r}\frac{(|\vec{n}|+\alpha_j)_{n_j}\left(\alpha_j-j+1+\sum_{i=1}^{j}n_i\right)_{n_j-1}}{(n_j-1)!}.$$

Proof. Theorem 2.1 gives us the Mellin transform of the left hand side; for the Mellin transform of the right hand side we write the $(r-1)$-fold differentiation as the $(r-1)$-fold application of a D_α-operator. This way we can use Lemma 2.6 and the fact that the Mellin transform of a Mellin convolution of functions is the product of the Mellin transform of these functions, and we obtain the desired result. □

Theorem 2.10. *The type II Jacobi-Piñeiro polynomials are given by the following formula*

$$P_{\vec{n}}(x) = K_{\vec{n}}^{II} x^{|\vec{n}|+r} \frac{d^{r-1}}{dx^{r-1}} x^{-|\vec{n}|-1} \\ \left(P_{n_1}^{(\alpha_1)}(x) * P_{n_2}^{(\alpha_2+n_1+1)}(x) x^{n_1+1} * \cdots * P_{n_r}^{(\alpha_r+n_1+n_2+\cdots+n_{r-1}+r-1)}(x) x^{n_1+n_2+\cdots+n_{r-1}+r-1}\right)$$

with

$$K_{\vec{n}}^{II} = (-1)^{r-1} \frac{|\vec{n}|!}{\prod_{j=1}^r n_j!} \prod_{j=1}^r \frac{\left(\alpha_j + \sum_{i=1}^j n_i + j\right)_{n_j}}{(\alpha_j + |\vec{n}| + 1)_{n_j}}.$$

Proof. The proof is analogous to the proof of Theorem 2.9. □

3 Irrationality of $\zeta(3)$

In 1979 Roger Apéry [1] published a proof that $\zeta(3) = \sum_{k=1}^\infty 1/k^3$ is irrational. Later, Beukers [4] showed that Apéry's construction is related to Hermite-Padé approximation to three functions, and as such it uses multiple orthogonal polynomials [15]. In this section it will be shown that this proof uses Hermite-Padé approximation (a mixture of type I and type II) with Jacobi-Piñeiro polynomials, and the Mellin transform is useful to find the explicit polynomials in this rational approximation problem.

The main idea for this irrationality proof is to construct a sequence of rational numbers that approximate $\zeta(3)$ better than possible if one would assume $\zeta(3)$ to be rational. This approach is summarized in the following lemma.

Lemma 3.1. *Let x be a real number. Suppose that one can find integer sequences p_n and q_n $(n \in \mathbb{N})$ such that*

1. *$p_n x - q_n \neq 0$ for all $n \in \mathbb{N}$.*
2. *$\lim_{n\to\infty}(p_n x - q_n) = 0$.*

Then x is irrational.

The proof of this lemma is very simple, but for those that really want to see a proof we refer to [15]. We will now construct rational approximations to certain functions f_1, f_2, f_3. Our rational approximants to $\zeta(3)$ will be the special case when we look at what happens for $f_3(1)$.

We will use three measures on $[0,1]$ with densities

$$w_1(x) = 1, \quad w_2(x) = -\log(x), \quad w_3(x) = \frac{1}{2}\log^2 x, \qquad x \in [0,1].$$

Their Stieltjes transforms

$$f_1(z) = \int_0^1 \frac{dx}{z-x}, \quad f_2(z) = -\int_0^1 \log x \frac{dx}{z-x}, \quad f_3(z) = \frac{1}{2}\int_0^1 \log^2 x \frac{dx}{z-x}$$

are analytic functions in $\mathbb{C} \setminus [0,1]$, and for $z \to 1$ one has $f_3(1) = \zeta(3)$. We are interested in a simultaneous rational approximation problem for the three functions f_1, f_2, f_3 near infinity of the following kind: find polynomials A_n, B_n of degree n and polynomials C_n, D_n such that

$$\begin{aligned} A_n(1) &= 0, && (3.1)\\ A_n(z)f_1(z) - B_n(z)f_2(z) - C_n(z) &= \mathcal{O}\left(\frac{1}{z^{n+1}}\right), \qquad z \to \infty, && (3.2)\\ A_n(z)f_2(z) - 2B_n(z)f_3(z) - D_n(z) &= \mathcal{O}\left(\frac{1}{z^{n+1}}\right), \qquad z \to \infty. && (3.3) \end{aligned}$$

Equation (3.2) is a type I Hermite-Padé approximation problem for the functions f_1, f_2 and (3.3) is a type I Hermite-Padé problem for f_2, f_3. Since both problems use the same polynomials A_n, B_n, the two equations (3.2)–(3.3) are a type II Hermite-Padé problem with common denominator (A_n, B_n). The extra condition (3.1) will be needed since we are interested only in $f_3(1)$ and not in $f_1(1)$ (which is divergent) and $f_2(1)$. The solution of such a Hermite-Padé problem always uses multiple orthogonal polynomials: let A_n, B_n be such that

$$\begin{aligned} \int_0^1 [A_n(x) + B_n(x)\log x]x^k\,dx &= 0, \qquad k = 0,1,\ldots,n-1, && (3.4)\\ \int_0^1 [A_n(x) + B_n(x)\log x]x^k \log x\,dx &= 0, \qquad k = 0,1,\ldots,n-1, && (3.5) \end{aligned}$$

then

$$\begin{aligned} C_n(z) &= \int_0^1 \left(\frac{A_n(z) - A_n(x)}{z-x} + \frac{B_n(z) - B_n(x)}{z-x}\log(x)\right) dx\\ D_n(z) &= \int_0^1 \left(\frac{A_n(z) - A_n(x)}{z-x} + \frac{B_n(z) - B_n(x)}{z-x}\log(x)\right) \log x\; dx \end{aligned}$$

are polynomials of degree $n-1$ and

$$\begin{aligned} A_n(z)f_1(z) - B_n(z)f_2(z) - C_n(z) &= \int_0^1 \frac{A_n(x) + B_n(x)\log x}{z-x}\,dx,\\ A_n(z)f_2(z) - 2B_n(z)f_3(z) - D_n(z) &= \int_0^1 \frac{A_n(x) + B_n(x)\log x}{z-x}\log x\,dx. \end{aligned}$$

The orthogonality conditions and the expansion of $1/(z-x)$ near $z = \infty$ then give the required behaviour near infinity in (3.2)–(3.3). Lemma 2.5 shows that (3.1) holds, and hence all the conditions for the proposed rational approximation problem are satisfied and the polynomials A_n, B_n can be obtained up to a common multiplicative factor. We can use Mellin transforms to find these polynomials explicitly. Let

$$F_n(x) = A_n(x) + B_n(x)\log(x), \qquad x \in [0,1],$$

Then the Mellin transform of F_n is given by

$$\widehat{F}_n(s) = \int_0^1 ([A_n(x) + B_n(x)\log x]x^s\,dx = C_n \frac{(-s)_n^2}{(s+1)_{n+1}^2}.$$

Indeed, the double poles at $-1, -2, \ldots, -n-1$ appear because F_n contains powers x^k and $x^k \log x$ for $k = 0, 1, \ldots, n$ as in Theorem 2.3. The double zeros at $0, 1, \ldots, n-1$ appear because of the orthogonality conditions (3.4)–(3.5). The Mellin transform is $\mathcal{O}(1/s^2)$ as $s \to \infty$ which, by Lemma 2.5, implies that $F_n(1) = A_n(1) = 0$. Hence everything is determined up to a constant multiplicative factor C_n. Now that $\widehat{F}_n$ is known, one can obtain the required properties of F_n that are needed to prove the irrationality of $\zeta(3)$, see [4, ?, 16]

4 Infinitely many $\zeta(2n+1)$ are irrational

At present $\zeta(3)$ is the only zeta value in an odd argument known to be irrational. However, Ball and Rivoal [3] proved that infinitely many $\zeta(2n+1)$ are irrational. Using the Mellin transform, their construction can be interpreted as the result of a Hermite-Padé approximation problem. To deduce this result from the approximation problem, we need Nesterenko's criterion for linear independence [10]

Lemma 4.1. *Let $\theta_1, \ldots, \theta_r$ be r real numbers. Suppose there exist r sequences of integers $(p_{k,n} : n \in \mathbb{N})$, $k = 1, 2, \ldots, r$, such that*

1. *There exist $0 < \alpha_1 \le \alpha_2 < 1$ such that*

$$\alpha_1^{n+o(n)} \le \left| \sum_{k=1}^{r} p_{k,n}\theta_k \right| \le \alpha_2^{n+o(n)}.$$

2. *There exists $\beta > 1$ such that for all $1 \le k \le r$ one has $|p_{k,n}| \le \beta^{n+o(n)}$.*

Then the dimension of the space spanned by $\theta_1, \theta_2, \ldots, \theta_r$ over $\mathbb{Q}$ is at least $(\log\beta - \log\alpha_1)/(\log\beta - \log\alpha_1 + \log\alpha_2)$.

We will use r measures on $[0,1]$ with densities

$$w_j(x) = \frac{(-1)^{j-1}}{(j-1)!}\log^{j-1}(x) \qquad j = 1, ..., r.$$

As in Section 3, their Stieltjes transforms

$$f_j(z) = \frac{(-1)^{j-1}}{(j-1)!}\int_0^1 \log^{j-1}(x)\frac{\mathrm{d}x}{z-x}$$

are analytic functions in $\mathbb{C}\backslash[0,1]$ and for $z \to 1$ one has $f_j(1) = \zeta(j)$ for $2 \le j \le r$.

We are interested in r polynomials $A_{n,j}$ of degree n and a polynomial B_n of degree $n-1$ that solve the following Hermite-Padé approximation problem, which is a mixture of type I and type II.

$$\sum_{j=1}^{r}(j-1)!(-1)^{j-1}A_{n,j}(z)f_j(z) - B_n(z) = \mathcal{O}\left(1/z^{\ell n+1}\right), \qquad z\to\infty \tag{4.1}$$

$$\sum_{j=1}^{r}(j-1)!A_{n,j}(z)f_j(1/z)/z + B_n^*(z) = \mathcal{O}\left(z^{(\ell+1)n}\right), \qquad z\to 0 \tag{4.2}$$

$$\sum_{j=1}^{r}A_{n,j}(x)\log^{j-1}(x) = \mathcal{O}\left((1-x)^{r(n+1)-2\ell n-1}\right), \quad x\to 1 \tag{4.3}$$

In this problem, $\ell < r/2$ and $B_n^*(z) = z^{n-1}B_n(1/z)$. Moreover, by comparing the number of coefficients to be determined with the number of conditions, we conclude that this problem will have a unique solution up to a constant factor. The condition (4.1) is typical for approximating zeta values. However, since we only want a combination of odd zeta values, we need some symmetry that cancels the even zeta values. This symmetry implies the extra condition at $z=0$ in (4.2). The introduction of ℓ allows us to place an additional number of interpolation conditions at $x=1$, which is done in (4.3).

Finding explicit expressions for the polynomials can be done more easily again, using the Mellin transform. If we define

$$F_n(z) = \sum_{j=1}^{r}A_{n,j}(z)\log^{j-1}z \tag{4.4}$$

then the orthogonality conditions following from (4.1) are equivalent with the condition that $\widehat{F_n}(s)$ has zeros at $s = 0, 1, ..., \ell n$. The symmetry imposed on the polynomials $A_{n,j}$ now results in symmetry on the zeros of $\widehat{F_n}(s)$, namely the orthogonality conditions following from (4.2) are equivalent with the condition that $\widehat{F_n}(s)$ has zeros at $s = -\ell n - n - 1, -\ell n - n, ..., -n - 2$.

Looking at (4.4), we see that $F_n(x)$ consists of terms with $x^k\log^j(x)$ for $0\le k\le n$ and $0\le j\le r-1$, so the poles of $\widehat{F_n}(s)$ should be $-n-1, -n, ..., -1$, all with multiplicity r. Finally, Lemma 2.5 assures us that the degree of the numerator of $\widehat{F_n}(s)$ can be no higher than $2\ell n$, hence we obtain

$$\widehat{F_n}(s) = \int_0^1 F_n(x)x^s dx = C_n\frac{(-s)_{\ell n}(s+n+2)_{\ell n}}{(s+1)_{n+1}^r}$$

and again everything is determined up to the constant factor C_n.

Letting r tend to infinity, the dimension of the $\mathbb{Q}$-vector space spanned by $\{\zeta(3), \zeta(5), \ldots\}$ also tends to infinity, giving the desired result. It is worth remarking that the authors also used these same approximations, but now with fixed $r = 167$ and $\ell = 10$, to prove that there exists an odd integer j with $5\le j\le 169$ such that $1, \zeta(3), \zeta(j)$ are $\mathbb{Q}$-linearly independent.

5 One of the numbers $\zeta(5), \zeta(7), \zeta(9), \ldots, \zeta(21)$ is irrational

This result was originally due to Rivoal, but was subsequently (and in two stages) improved by Zudilin [19], [20] who proved that one of the numbers $\zeta(5), \zeta(7), \zeta(9), \zeta(11)$ is

irrational. The approach used in these papers can be put in the Hermite-Padé framework: we will present an approximation problem that gives as a solution exactly the construction used by Rivoal.

In this section we will use a generalization of Lemma 3.1.

Lemma 5.1. *Let $x_1, x_2, \ldots, x_r$ be real numbers. Suppose there exist $r+1$ integer sequences $(p_{n,k} : n \in \mathbb{N})$, $k = 0, 1, 2, \ldots, r$ such that*

1. $\sum_{k=1}^{r} p_{n,k}x_k - p_{n,0} \neq 0$ *for all* $n \in \mathbb{N}$.
2. $\lim_{n\to\infty} (\sum_{k=1}^{r} p_{n,k}x_k - p_{n,0}) = 0$.

Then at least one of the numbers $x_1, x_2, \ldots, x_r$ is irrational.

The proof is a simple extension of the proof of Lemma 3.1, or this lemma can be seen as a special case of Nesterenko's criterion (Lemma 4.1). The rational approximation problem will involve two-point (at 0 and ∞) mixed type I,II Hermite-Padé approximation to functions $f_1, f_2, \ldots, f_r, f_{r+1}, f_{r+2}$ of the form

$$f_j(z) = \frac{(-1)^{j-1}}{(j-1)!} \int_0^1 \log^{j-1} x \frac{dx}{z-x},$$

with extra interpolation conditions at 1 and some symmetry. Indeed, we want to find polynomials $A_{n,1}, A_{n,2}, \ldots, A_{n,r}$ of degree at most n and B_n, C_n, D_n of degree at most $n-1$, such that near infinity

$$\sum_{j=1}^{r} (j-1)!(-1)^{j-1} A_{n,j}(z) f_j(z) - B_n(z) = \mathcal{O}(1/z^{n+1}), \qquad z \to \infty \tag{5.1}$$

$$\sum_{j=1}^{r} j!(-1)^{j} A_{n,j}(z) f_{j+1}(z) - C_n(z) = \mathcal{O}(1/z^{n+1}), \qquad z \to \infty \tag{5.2}$$

$$\sum_{j=1}^{r} (j+1)!(-1)^{j+1} A_{n,j}(z) f_{j+2}(z) - D_n(z) = \mathcal{O}(1/z^{n+1}), \qquad z \to \infty \tag{5.3}$$

and near 0

$$\sum_{j=1}^{r} -(j-1)! A_{n,j}(z) f_j(1/z)/z + B_n^*(z) = \mathcal{O}(z^{2n}), \qquad z \to 0 \tag{5.4}$$

$$\sum_{j=1}^{r} j! A_{n,j}(z) f_{j+1}(1/z)/z + C_n^*(z) = \mathcal{O}(z^{2n}), \qquad z \to 0 \tag{5.5}$$

$$\sum_{j=1}^{r} -(j+1)! A_{n,j}(z) f_{j+2}(1/z)/z + D_n^*(z) = \mathcal{O}(z^{2n}), \qquad z \to 0 \tag{5.6}$$

and near 1

$$\sum_{k=1}^{r} A_{n,k}(x) \log^{k-1} x = \mathcal{O}((x-1)^{r(n+1)-6n-2}), \qquad x \to 1, \tag{5.7}$$

and we require the symmetry property

$$A^*_{n,j}(x) = (-1)^{j+1} A_{n,j}(x), \tag{5.8}$$

where $A^*_{n,j}(x) = x^n A_{n,j}(1/x)$. The latter condition actually imposes a lot of conditions on the coefficients of $A_{n,j}$. However, the condition (5.8) with the conditions (5.1)-(5.3) imply the conditions (5.4)-(5.6).

Observe that

$$f_j(1/z)/z = \frac{(-1)^{j-1}}{(j-1)!} \int_0^1 \log^{j-1} x \frac{dx}{1-zx} = \sum_{n=0}^{\infty} \frac{z^n}{(n+1)^j}$$

so that $f_j(1/z)/z$ is analytic in $|z| < 1$.

As in Section 4, the second set of conditions (5.4)-(5.6) results from the first set (5.1)-(5.3), using the symmetry (5.8) which is needed to get rid of the even zeta values. Moreover, each set consists of three conditions: this is necessary to get rid of $\zeta(3)$. Putting only one condition instead of three would give the irrationality of at least one number in a finite set $\{\zeta(3), \zeta(5), \ldots\}$, which would not be an interesting result because of the already established irrationality of $\zeta(3)$. Equation (5.7) consists of additional interpolation conditions near 1.

Obtaining the Mellin transform $\widehat{F_n}(s)$ is done in the same way as in Section 4. Note that here the triple conditions at 0 and ∞ imply that not only $\widehat{F_n}$ has zeros at $0, 1, \ldots, n-1$ and at $-n-2, -n-3, \ldots, -2n-1$, but also its first and second derivatives have the same zeros. This leads us to factors $(-s)_n^3 (s+n+2)_n^3$ in the numerator of $\widehat{F_n}(s)$. The structure of $F_n(x)$ implies r-fold poles at $-1, -2, \ldots, -n-1$. Lemma 2.5 allows only one more linear factor in the numerator of $\widehat{F_n}(s)$, which, according to (5.8), can only be $(s + n/2 + 1)$ (up to a constant). So we obtain

$$\widehat{F_n}(s) = \int_0^1 F_n(x) x^s dx = C_n \left(s + \frac{n}{2} + 1\right) \frac{(-s)_n^3 (s+n+2)_n^3}{(s+1)_{n+1}^r}.$$

and once again everything is determined up to the constant factor C_n.

6 q-Mellin transforms for multiple little q-Jacobi polynomials

We will now consider multiple little q-Jacobi polynomials, i.e., we consider weights on the exponential lattice $\{q^k, k \in \mathbb{N}\}$

$$d\mu_j(x) = w_j(x)\, d_q x, \qquad j = 1, \ldots, r,$$

with $w_j(x) = x^{\alpha_j} \frac{(qx;q)_\infty}{(q^{\beta+1}x;q)_\infty}$, $j = 1, 2, \ldots, r$, where $\alpha_j - \alpha_i \notin \mathbb{Z}$. The basis q is such that $0 < |q| < 1$, and in the irrationality applications further restrictions on q will be imposed. We will focus on the case $\beta = 0$ so that $w_j(x) = x^{\alpha_j}$. Their q-Mellin transforms turn out to be rational functions. The q-Mellin transform $\hat{f}(s)$ of a measurable function f on the q-exponential lattice on $(0, 1]$ is given by

$$\hat{f}(s) = \int_0^1 f(x) x^s \, d_q x, \tag{6.1}$$

where the q-integration means

$$\int_0^1 f(x) d_q x = \sum_{k=0}^{\infty} q^k f(q^k) \tag{6.2}$$

and this transform is defined for any s for which the series $\sum_{k=0}^{\infty} q^{k(s+1)} |f(q^k)|$ is convergent.

Theorem 6.1. *Let $(A_{\vec{n},1}, \ldots, A_{\vec{n},r})$ be the type I multiple orthogonal polynomials for multiple little q-Jacobi weights with parameters $(\alpha_1, \ldots, \alpha_r; \beta = 0)$. Then*

$$\int_0^1 \left(\sum_{j=1}^{r} A_{\vec{n},j}(x) x^{\alpha_j} \right) x^s \, d_q x = C^I_{\vec{n},q} \, \frac{(q^s; q^{-1})_{|\vec{n}|-1}}{\prod_{j=1}^{r} (q^{s+\alpha_j+1}; q)_{n_j}}, \qquad \Re s \geq 0, \tag{6.3}$$

where $(a;q)_n = (1-a)(1-aq)\,(1-aq^2)\cdots(1-aq^{n-1})$ are the q-Pochhammer symbols and

$$C^I_{\vec{n},q} = \frac{\prod_{j=1}^{r} (q^{|\vec{n}|+\alpha_j}; q)_{n_j}}{(q;q)_{|\vec{n}|-1}}. \tag{6.4}$$

Proof. Let us denote the polynomial $A_{\vec{n},j}$ by

$$A_{\vec{n},j}(x) = \sum_{\ell=0}^{n_j-1} a_\ell(\vec{n}, j) x^\ell,$$

then for $\Re s \geq 0$

$$\int_0^1 A_{\vec{n},j}(x) x^{\alpha_j} x^s \, d_q x = \sum_{\ell=0}^{n_j-1} \frac{a_\ell(\vec{n}, j)}{1 - q^{s+\ell+\alpha_j+1}} = \frac{\pi_{n_j-1}(q^s; \vec{n}, j)}{(q^{s+\alpha_j+1}; q)_{n_j}},$$

where π_{n_j-1} is some polynomial of degree $n_j - 1$. This means that

$$\begin{aligned} \int_0^1 \left(\sum_{j=1}^{r} A_{\vec{n},j}(x) x^{\alpha_j} \right) x^s \, d_q x &= \sum_{j=1}^{r} \frac{\pi_{n_j-1}(q^s; \vec{n}, j)}{(q^{s+\alpha_j+1}; q)_{n_j}} \\ &= \frac{\pi_{|\vec{n}|-1}(s)}{(q^{s+\alpha_1+1}; q)_{n_1} (q^{s+\alpha_2+1}; q)_{n_2} \cdots (q^{s+\alpha_r+1}; q)_{n_r}}, \end{aligned}$$

where $\pi_{|\vec{n}|-1}$ is some polynomial of degree $|\vec{n}|-1$. The orthogonality conditions (1.1) show that this q-Mellin transform is zero for $s = 0, 1, \ldots, |\vec{n}|-2$, which gives $|\vec{n}|-1$ zeros of the polynomial $\pi_{|\vec{n}|-1}$, and hence $\pi_{|\vec{n}|-1}(s) = C(1-q^s)(1-q^{s-1})(1-qs-2)\cdots(1-q^{s-|\vec{n}|+2}) = (q^s; q^{-1})_{|\vec{n}|-1}$, with C a constant factor. The normalizing condition (1.2) shows that the q-Mellin transform at $s = |\vec{n}| - 1$ is equal to 1, from which the value of the constant $C^I_{\vec{n},q}$ can be determined. □

Observe that the q-Mellin transform of the appropriate combination of type I little q-Jacobi polynomials is a rational function in q^s for which

- The poles are all simple and form r arithmetic sequences $s = \{-\alpha_j - k : \ k = 1, 2, \ldots, n_j\}$, $(1 \leq j \leq r)$. Hence once the poles are known, one can determine the parameters $\alpha_1, \ldots, \alpha_r$ and the multi-index $\vec{n} = (n_1, \ldots, n_r)$.

- The zeros are at non-negative integers: $s = 0, 1, 2, \ldots, |\vec{n}| - 2$ and they give the appropriate orthogonality conditions.

The q-Mellin transform of the type II multiple little q-Jacobi polynomials is given by the following theorem

Theorem 6.2. *Let $P_{\vec{n}}$ be the type II multiple orthogonal polynomial for the multiple little q-Jacobi weights with parameters $(\alpha_1, \ldots, \alpha_r; \beta = 0)$. Then*

$$\int_0^1 P_{\vec{n}}(x) x^s \, d_q x = C^{II}_{\vec{n},q} \, \frac{\prod_{j=1}^r (q^{s-\alpha_j}; q^{-1})_{n_j}}{(q^{s+1}; q)_{|\vec{n}|+1}}, \qquad \Re s > 0, \tag{6.5}$$

with

$$C^{II}_{\vec{n},q} = \frac{(q^{-1}; q^{-1})_{|\vec{n}|}}{\prod_{j=1}^r \left(q^{-|\vec{n}|-\alpha_j-1}; q^{-1}\right)_{n_j}} = q^{\sum_{1\le j\le k\le r} n_j n_k + \sum_{j=1}^r n_j \alpha_j} \frac{(q;q)_{|\vec{n}|}}{\prod_{j=1}^r (q^{|\vec{n}|+\alpha_j+1}; q)_{n_j}}. \tag{6.6}$$

Proof. Let $P_{\vec{n}}(x) = \sum_{k=0}^{|\vec{n}|} b_k(\vec{n}) x^k$, where $b_{|\vec{n}|}(\vec{n}) = 1$, then

$$\int_0^1 P_{\vec{n}}(x) x^s \, d_q x = \sum_{k=0}^{|\vec{n}|} \frac{b_k(\vec{n})}{1 - q^{k+s+1}} = \frac{\pi_{|\vec{n}|}(q^s)}{(q^{s+1}; q)_{|\vec{n}|+1}},$$

where $\pi_{|\vec{n}|}$ is some polynomial of degree $|\vec{n}|$. The orthogonality conditions (1.4) show that this q-Mellin transform is zero for $s = \alpha_j + k$, $k = 0, 1, \ldots, n_j - 1$ and $1 \le j \le r$, which gives $|\vec{n}|$ zeros for $\pi_{|\vec{n}|}$ and hence $\pi_{|\vec{n}|}(q^s) = C \prod_{j=1}^r (q^{s-\alpha_j}; q^{-1})_{n_j}$. We have normalized $P_{\vec{n}}$ to be a monic polynomial, which makes $b_{|\vec{n}|}(\vec{n}) = 1$, and since this is the residue at $s = -|\vec{n}| - 1$, we can determine the constant factor $C^{II}_{\vec{n},q}$ by

$$1 = \lim_{s \to -|\vec{n}|-1} \left(1 - q^{s+|\vec{n}|+1}\right) \frac{\pi_{|\vec{n}|}(q^s)}{(q^{s+1}; q)_{|\vec{n}|+1}},$$

which gives the constant in (6.6). □

Observe that the q-Mellin transform of type II little q-Jacobi polynomials is also a rational function in q^s:

- The poles are found for s a negative integer: $s = -1, -2, \ldots, -|\vec{n}| - 1$, from which we can determine the length $|\vec{n}|$ of the multi-index.
- The zeros are simple and form r arithmetic sequences $s = \{\alpha_j + k : \ k = 0, \ldots, n_j - 1\}$ for $1 \le j \le r$. This allows us to determine the parameters $\alpha_1, \ldots, \alpha_r$ and the multi-index $\vec{n} = (n_1, \ldots, n_r)$. These zeros give the appropriate orthogonality conditions.

The roles of the zeros and poles are interchanged for type I and type II polynomials.

6.1 Multiple zeros or poles

If $\alpha_i - \alpha_j \notin \mathbb{Z}$ then all the poles (for type I) and the zeros (for type II) of the q-Mellin transform of little q-Jacobi polynomials are simple. Multiple zeros or poles appear in the confluent case when $\alpha_i - \alpha_j \to 0$. The weight functions x^{α_i} and x^{α_j} on $[0,1]$ then have to be replaced by x^{α_i} and $-\log_q(x)x^{\alpha_i}$ in order to have enough independent orthogonality conditions. The following result holds

Theorem 6.3. *Let $(A_{\vec{n},1}, \ldots, A_{\vec{n},r})$ be the type I multiple orthogonal polynomials for the weights*

$$\left(x^\alpha, -\log_q x \; x^\alpha, \ldots, (-1)^{r-1} \frac{\log_q^{r-1} x}{(r-1)!} \; x^\alpha \right), \qquad x \in [0,1],$$

where $\alpha > -1$. Then for $n_1 \geq n_2 \geq \cdots \geq n_r$

$$\int_0^1 \left(\sum_{j=1}^r A_{\vec{n},j}(x)(-1)^{j-1} \frac{\log_q^{j-1} x}{(j-1)!} \; x^\alpha \right) x^s \, d_q x = C^I_{\vec{n},q} \frac{(q^s;q^{-1})_{|\vec{n}|-1}}{\prod_{j=1}^r (q^{s+\alpha+1};q)_{n_j}}.$$

Proof. Consider the q-Mellin transform

$$M_k(s) = (-1)^k \int_0^1 x^s \frac{\log_q^k x}{k!} \, d_q x,$$

then it is an easy proof by induction to show that $M_k(s) = \frac{\pi_k(q^s)}{(1-q^{s+1})^{k+1}}$. Hence, for $A_{\vec{n},j}(x) = \sum_{\ell=0}^{n_j-1} a_\ell(\vec{n},j) x^\ell$ we then find

$$\int_0^1 (-1)^{j-1} A_{\vec{n},j}(x) x^\alpha \frac{\log_q^{j-1} x}{(j-1)!} x^s \, d_q x = \sum_{\ell=0}^{n_j-1} \frac{a_\ell(\vec{n},j)\pi_{j-1}(q^s)}{(1-q^{s+\alpha+\ell+1})^j} = \frac{\pi'_{n_j-1}(q^s)}{(q^{s+\alpha+1};q)_{n_j}^j},$$

where π_{j-1} and π'_{n_j-1} are polynomials of degree $j-1$ and n_j-1, respectively. Adding all such terms together gives

$$\int_0^1 \left(\sum_{j=1}^r A_{\vec{n},j}(x)(-1)^{j-1} \frac{\log_q^{j-1} x}{(j-1)!} \; x^\alpha \right) x^s \, d_q x = \frac{\pi_{|\vec{n}|-1}(q^s)}{\prod_{j=1}^r (q^{s+\alpha+1};q)_{n_j}},$$

where $\pi_{|\vec{n}|-1}$ is a polynomial of degree $|\vec{n}|-1$. The orthogonality conditions then show that this q-Mellin transform is zero for $s = 0, 1, \ldots, |\vec{n}|-2$, which gives $|\vec{n}|-1$ zeros for $\pi_{|\vec{n}|-1}$. The normalization determines the constant factor. □

The condition $n_1 \geq n_2 \geq \cdots \geq n_r$ is needed in order to make

$$1, x, \ldots, x^{n_1-1}, -\log_q x, (-\log_q x)x, \ldots, (-\log_q x)x^{n_2-1}, \ldots, \\ (-1)^{r-1} \frac{\log_q^{r-1} x}{(r-1)!}, \ldots, (-1)^{r-1} \frac{\log_q^{r-1} x}{(r-1)!} x^{n_r-1}$$

a Chebyshev system on $[0,1]$, so that there are enough independent orthogonality conditions for the multiple orthogonal polynomials.

Theorem 6.4. *Let $P_{\vec{n}}$ be the type II multiple orthogonal polynomial for the weights*

$$\left(x^\alpha, -\log_q x\ x^\alpha, \ldots, (-1)^{r-1}\frac{\log_q^{r-1} x}{(r-1)!}\ x^\alpha \right), \qquad x \in [0,1],$$

where $\alpha > -1$. Then for $n_1 \geq n_2 \geq \cdots \geq n_r$

$$\int_0^1 P_{\vec{n}}(x) x^s\, d_q x = C^{II}_{\vec{n},q} \frac{\prod_{j=1}^r (q^{s-\alpha};q^{-1})_{n_j}}{(q^{s+1};q)_{|\vec{n}|+1}}. \tag{6.7}$$

Proof. Let $P_{\vec{n}}(x) = \sum_{k=0}^{|\vec{n}|} b_k(\vec{n}) x^k$, where $b_{|\vec{n}|}(\vec{n}) = 1$, then

$$\int_0^1 P_{\vec{n}}(x) x^s\, d_q x = \sum_{k=0}^{|\vec{n}|} \frac{b_k(\vec{n})}{1-q^{s+k+1}} = \frac{\pi_{|\vec{n}|}(q^s)}{(q^{s+1};q)_{|\vec{n}|+1}},$$

where $\pi_{|\vec{n}|}$ is some polynomial of degree $|\vec{n}|$. If we take the kth derivative, then

$$\int_0^1 P_{\vec{n}}(x) x^s \left(\log^k x\right)\, d_q x = \frac{d^k}{ds^k}\left(\frac{\pi_{|\vec{n}|}(q^s)}{(q^{s+1};q)_{|\vec{n}|+1}} \right).$$

The orthogonality conditions then show that $\pi_{|\vec{n}|}$ has zeros at $s = \alpha, \alpha+1, \ldots, \alpha+n_1-1$, the first n_2 of these zeros have at least multiplicity two, the first n_3 of these zeros have at least multiplicity three, etc. This gives all the zeros of $\pi_{|\vec{n}|}$ with their appropriate multiplicity. The normalizing condition gives the correct value of the multiplicative factor. □

6.2 Rodrigues formula

The advantage of working with the q-Mellin transform of multiple little q-Jacobi polynomials is that it is quite simple and contains the parameters and the multi-index in a very explicit way. One can derive several properties of multiple little q-Jacobi polynomials from this q-Mellin transforms by using properties of the q-Mellin transform. The q-Mellin convolution $f * g$ of two functions f and g on $[0,1]$ is defined as

$$(f*g)(x) = \int_{qx}^1 f(t) g(x/t) \frac{d_q t}{t}, \qquad x \in [0,1],$$

and it has the property that the q-Mellin transform of $f*g$ is the product of the q-Mellin transforms $\hat{f}$ and $\hat{g}$. Hence if we can recognize a q-Mellin transform as a product of two other q-Mellin transforms, then the original function is a q-Mellin convolution. Take for instance the q-Mellin transform in (6.3): this is clearly the product of r functions multiplied with an extra polynomial of degree $r-1$:

$$\frac{(q^s;q^{-1})_{|\vec{n}|-1}}{\prod_{j=1}^r (q^{s+\alpha_j+1};q)_{n_j}} = \left(\prod_{j=1}^r \frac{(q^{s-\sum_{i=1}^{j-1} n_i};q^{-1})_{n_j-1}}{(q^{s+\alpha_j+1};q)_{n_j}} \right) \left(\prod_{j=1}^{r-1} (1-q^{s+1-\sum_{i=1}^j n_i}) \right).$$

Each factor can be recognized as the q-Mellin transform of a little q-Jacobi polynomial:

$$\frac{(q^{s-\sum_{i=1}^{j-1} n_i};q^{-1})_{n_j}}{(q^{s+\alpha_j+1};q)_{n_j+1}} = \int_0^1 P_{n_j}^{(\alpha_j+\sum_{i=1}^{j-1} n_i, 0)}(x) x^{\alpha_j} x^s\, d_q x,$$

where $P_n^{(\alpha,\beta)}$ is the little q-Jacobi polynomial on $[0,1]$ with weight $x^\alpha \frac{(qx;q)_\infty}{(q^{\beta+1}x;q)_\infty}$ and with a suitable normalization.

We will now give a number of lemmas that will lead us to a Rodrigues formula for type II multiple little q-Jacobi polynomials, and to an explicit expression of both type I and II polynomials in terms of the (simple) little q-Jacobi polynomials. In the proofs of these lemmas, we will often make use of the concept of 'summation by parts', which can be interpreted as a q-analogue of integration by parts. In this context we will use the notation $p = q^{-1}$. The q-analogues of the derivative which are needed in this summation by parts technique, are given by

$$\mathcal{D}_q f(x) = \frac{f(x) - f(qx)}{x - qx}, \qquad \mathcal{D}_p f(x) = \frac{f(x) - f(px)}{x - px}.$$

Lemma 6.5. *If $g(p) = 0$ or $f(1) = 0$ then*

$$\int_0^1 f(x)\mathcal{D}_p g(x) d_q x = -q \int_0^1 g(x)\mathcal{D}_q f(x) d_q x.$$

Proof. This is straightforward if one uses the definitions of q-integration and q-differentiation mentioned above. □

Lemma 6.6. *If $f(1) = \mathcal{D}_q f(1) = \ldots = \mathcal{D}_q^{(k-1)} f(1) = 0$ and*

$$\hat{f}(s) = \int_0^1 f(x) x^s \, d_q x,$$

then

$$\lim_{s\to\infty} (q^{-s-1};q^{-1})_k \hat{f}(s) = \begin{cases} 0 & \text{if } k = 0, \\ (1-q)^k \mathcal{D}_q^{(k)} f(1) & \text{if } k > 0. \end{cases}$$

Proof. For $k = 0$ we write the q-Mellin transform $\hat{f}(s)$ as an infinite series, using the definition of q-integration (6.2). Taking the limit for s to infinity inside this sum, we see that every term vanishes. For $k > 0$ we use k times summation by parts on $\hat{f}(s) = \int_0^1 f(x)x^s d_q x$ (Lemma 6.5) in which we use that $\mathcal{D}_p \left(x^{s+1} \frac{1-p}{1-p^{s+1}} \right) = x^s$. This gives us

$$\hat{f}(s) = \frac{(1-q)^k}{(q^{-s-1};q)_k} \int_0^1 x^{s+k} \mathcal{D}_q^{(k)} f(x) d_q x.$$

Writing the q-integral as an infinite sum, we see that taking the limit for s to infinity, every term (but one) vanishes, and the result follows. □

Lemma 6.7. *Suppose $f(q^{-1}) = 0$, then*

$$-q \frac{1 - q^{s-\alpha}}{1-q} \hat{f}(s) = \widehat{D_\alpha^* f}(s),$$

where $(D_\alpha^* f)(x) = x^{-\alpha} \mathcal{D}_{q^{-1}} x^{\alpha+1} f(x)$.

Proof. Summation by parts gives

$$\int_0^1 x^{s-\alpha} \left(\mathcal{D}_{q^{-1}} [x^{\alpha+1} f(x)] \right) d_q x = -q \frac{1 - q^{s-\alpha}}{1-q} \int_0^1 x^{\alpha+1} f(x) x^{-\alpha+s-1} \, d_q x$$

which is the desired result. □

Lemma 6.8. *The q-Mellin transform of the density $(qx;q)_n$ on $[0,1]$ is given by*

$$\int_0^1 (qx;q)_n x^s \, d_q x = \frac{(q;q)_n}{(q^{s+1};q)_{n+1}}, \qquad \Re s > -1.$$

Proof. Repeated (n times) summation by parts on the expression $(qx;q)_n x^s$ will lower the index of the q-Pochhammer symbol, and raise the power of x. The fact that $\int_0^1 x^{n+s} d_q x = \frac{1}{1-q^{n+s+1}}$ then immediately gives the statement in the lemma. □

Theorem 6.9. *The type II multiple little q-Jacobi polynomials are given by the following Rodrigues formula*

$$P_{\vec{n}}(x) = \frac{C^{II}_{\vec{n},q}(1-q^{-1})^{|\vec{n}|}}{(q;q)_{|\vec{n}|}} \prod_{j=1}^{r} \left(x^{-\alpha_j} \mathcal{D}_{q^{-1}}{}^{n_j} x^{\alpha_j+n_j} \right) (qx;q)_{|\vec{n}|},$$

where the product of the r differential operators can be taken in any order.

Proof. Using Lemma 6.8, the q-Mellin transform in (6.5) can be written as

$$\widehat{P_{\vec{n}}}(s) = C^{II}_{\vec{n},q} \prod_{j=1}^{r} \left(q^{s-\alpha_j}; q^{-1} \right)_{n_j} \ \widehat{f_{|\vec{n}|}}(s),$$

with

$$\widehat{f_{|\vec{n}|}}(s) = \frac{1}{(q^{s+1};q)_{|\vec{n}|+1}} = \frac{\widehat{(qx;q)_{|\vec{n}|}}(s)}{(q;q)_{|\vec{n}|}}$$

and hence by Lemma 6.7 we find that $(q^{s-\alpha_r};q^{-1})_{n_r} \widehat{f_{|\vec{n}|}}(s)$ is the q-Mellin transform of

$$\frac{(1-q^{-1})^{n_r}}{(q;q)_{|\vec{n}|}} D^*_{\alpha_r} D^*_{\alpha_r+1} \cdots D^*_{\alpha_r+n_r-1} (qx;q)_{|\vec{n}|}.$$

The product of the n_r differential operators simplifies to

$$D^*_{\alpha_r} D^*_{\alpha_r+1} \cdots D^*_{\alpha_r+n_r-1} = x^{-\alpha_r} \mathcal{D}^{n_r}_{q^{-1}} x^{\alpha_r+n_r}.$$

If we repeat this procedure, then the Rodrigues formula follows. The product of the differential operators can be taken in any order since the product of the polynomial factors in the q-Mellin transform is commutative. □

Not only do we have this Rodrigues formula which gives an explicit expression for type II multiple little q-Jacobi polynomials, we can also express type I and II multiple little q-Jacobi polynomials as a convolution of the (simple) orthogonal little q-Jacobi polynomials. For these simple orthogonal polynomials we choose to normalize them so that they are monic: in this way they follow the normalization of type II multiple little q-Jacobi polynomials, hence we denote them by $P^{(\alpha)}_n$.

Theorem 6.10. *The type I multiple little q-Jacobi polynomials are given by the following formula*

$$\sum_{j=1}^{r} A_{\vec{n},j}(x) x^{\alpha_j} = C^{I}_{\vec{n},q} \left(\frac{1-q}{-q} \right)^{r-1} \prod_{j=1}^{r} \frac{\left(q^{-\alpha_j - \sum_{i=1}^{j} n_i + j - 1}; q^{-1} \right)_{n_j+1}}{(q^{-1};q^{-1})_{n_j-1}} x^{-|\vec{n}|+r} \mathcal{D}^{(r-1)}_{q^{-1}} x^{|\vec{n}|-1}$$
$$\left(P^{(\alpha_1)}_{n_1-1}(x) x^{\alpha_1} * P^{(\alpha_2+n_1-1)}_{n_2-1}(x) x^{\alpha_2} * \cdots * P^{(\alpha_r+n_1+n_2+\cdots+n_{r-1}-r+1)}_{n_r-1} x^{\alpha_r} \right).$$

Proof. Theorem 6.1 gives us the q-Mellin transform of the left hand side; for the q-Mellin transform of the right hand side we write the $(r-1)$-fold differentiation as the $(r-1)$-fold application of a D^*_α-operator. This way we can use Lemma 6.7 and the fact that the q-Mellin transform of a q-Mellin convolution of functions is the product of the q-Mellin transform of these functions, and we obtain the desired result. □

Theorem 6.11. *The type II multiple little q-Jacobi polynomials are given by the following formula*

$$P^{(\vec{\alpha})}_{\vec{n}}(x) = C^{II}_{\vec{n},q}\left(\frac{1-q}{-q}\right)^{r-1}\prod_{j=1}^{r}\frac{\left(q^{-\alpha_j-\sum_{i=1}^{j}n_i-j};q^{-1}\right)_{n_j}}{\left(q^{-1};q^{-1}\right)_{n_j}}x^{|\vec{n}|+r}\mathcal{D}^{(r-1)}_{q^{-1}}x^{-|\vec{n}|-1}$$
$$\left(P^{(\alpha_1)}_{n_1}(x) * P^{(\alpha_2+n_1+1)}_{n_2}(x)x^{n_1+1} * \cdots * P^{(\alpha_r+n_1+n_2+\cdots+n_{r-1}+r-1)}_{n_r}(x)x^{n_1+n_2+\cdots+n_{r-1}+r-1}\right).$$

Proof. The proof is analogous to the proof of Theorem 6.10. □

7 Is $\zeta_q(3)$ irrational?

When one tries to give a q-analogue (with $q = 1/p, p > 1, p \in \mathbb{N}$) of the irrationality proof of $\zeta(3)$, as is done in [9], the most logical approach would be to pose a mixed (type I, type II) Hermite-Padé approximation problem to the functions

$$f_1(z) = \int_0^1 \frac{d_qx}{z-x}, \qquad f_2(z) = \int_0^1 \frac{\log_q(x)d_qx}{z-x}, \qquad f_3(z) = \int_0^1 \frac{\log_q^2 x d_qx}{z-x}.$$

So we now want to find polynomials A_n, B_n of degree n and polynomials C_n, D_n of degree $n-1$, such that

$$A_n(1) = 0, \tag{7.1}$$
$$A_n(z)f_1(z) - B_n(z)f_2(z) - C_n(z) = \mathcal{O}\left(\frac{1}{z^{n+1}}\right), \qquad z\to\infty, \tag{7.2}$$
$$A_n(z)f_2(z) - 2B_n(z)f_3(z) - D_n(z) = \mathcal{O}\left(\frac{1}{z^{n+1}}\right), \qquad z\to\infty. \tag{7.3}$$

Introducing $F_n(x) = A_n(x) + B_n(x)\log_q(x)$, this gives rise to the orthogonality conditions

$$\int_0^1 F_n(x)x^k\, d_qx = 0, \qquad k = 0,1,\ldots,n-1, \tag{7.4}$$
$$\int_0^1 F_n(x)x^k\log_q x\, d_qx = 0, \qquad k = 0,1,\ldots,n-1. \tag{7.5}$$

The definition of $F_n(x)$ implies that the q-Mellin transform of $F_n(x)$ has $(q^{s+1};q)^2_{n+1}$ as a denominator; because of (7.3) its numerator should be a polynomial of degree $2n$ in q^s, and the orthogonality conditions (7.4-7.5) give us double zeros for $s = 0,1,\ldots,n-1$. Hence

$$\widehat{F_n}(s) = C_n\frac{(q^{s+1-n};q)^2_n}{(q^{s+1};q)^2_{n+1}},$$

where we can put the constant C_n equal to 1, since multiplying a solution F_n of the approximation problem with a factor, still gives a solution of the problem. Putting

$$S_n = \sum_{k=1}^{\infty} \frac{\mathrm{d}}{\mathrm{d}k} \left(q^k \widehat{F_n}(k-1) \right),$$

one can show that (see [9])

$$S_n = \ln(q)[a_n \zeta(3) - b_n].$$

a_n and b_n are rational numbers, and one can find a factor $\bar{D}_n$ such that $a_n\bar{D}_n, b_n\bar{D}_n$ are integers. Unfortunately, the sequence $\frac{S_n\bar{D}_n}{\ln(q)}$, which is then a sequence of linear combinations of 1 and $\zeta_q(3)$ with integer coefficients, does not tend to zero, so the expected irrationality of $\zeta_q(3)$ cannot be proved using this approximation.

8 $\mathbb{Q}$-linear independence of 1, $\zeta_q(1), \zeta_q(2)$

In [8] a type II Hermite-Padé approximation problem is considered that gives the linear independence over $\mathbb{Q}$ of the values $1, \zeta_q(1), \zeta_q(2)$. Here $\zeta_q(s)$ is a q-extension of the zeta value $\zeta(s)$. It is defined as

$$\zeta_q(s) = \sum_{k=1}^{\infty} \frac{k^{s-1}q^k}{1-q^k}, \qquad |q| < 1, s \in \mathbb{N}$$

and it is justified to call this a q-analogue of the zeta function since

$$\lim_{\substack{q \to 1 \\ |q|<1}} (1-q)^s \zeta_q(s) = (s-1)!\zeta(s).$$

The linear independence result holds for $q = 1/p$, with $p \in \mathbb{N}$. The Hermite-Padé approximation consists of approximations to two functions:

$$f_1(z) = \int_0^1 \frac{d_q x}{z-x} = \sum_{k=0}^{\infty} \frac{q^k}{z-q^k} \qquad \text{and} \qquad f_2(z) = \int_0^1 \frac{\log_q x d_q x}{z-x} = \sum_{k=0}^{\infty} \frac{kq^k}{z-q^k}.$$

The problem is to find polynomials $P_{n,m}$ of degree $\leq n+m$ and $Q_{n,m}, R_{n,m}$ of degree $\leq n+m-1$ such that

$$P_{n,m}(z)f_1(z) - Q_{n,m}(z) = \mathcal{O}\left(\frac{1}{z^{n+1}}\right), \qquad z \to \infty \tag{8.1}$$

$$P_{n,m}(z)f_2(z) - R_{n,m}(z) = \mathcal{O}\left(\frac{1}{z^{m+1}}\right), \qquad z \to \infty \tag{8.2}$$

where m, n are integers and $m \leq n$ is needed for normality. The orthogonality conditions that follow from this approximation problem are given by

$$\int_0^1 P_{n,m}(x)x^k \, d_q x = 0, \qquad k = 0, 1, \ldots, n-1, \tag{8.3}$$

$$\int_0^1 P_{n,m}(x)x^k \log_q x \, d_q x = 0, \qquad k = 0, 1, \ldots, m-1. \tag{8.4}$$

Since a term x^i has q-Mellin transform $\frac{1}{1-q^{i+s+1}}$, it is obvious that

$$\widehat{P_{n,m}}(s) = \frac{\Pi_{n+m}(q^s)}{(q^{s+1};q)_{n+m+1}}$$

with Π_{n+m} a polynomial of degree $n+m$. The first set of conditions gives n zeros at $0,1,\ldots,n-1$ of $\widehat{P_{n,m}}(s)$, the second set implies that the m zeros at $0,1,\ldots,m-1$ are double zeros. Hence

$$\widehat{P_{n,m}}(s) = C\frac{(q^s;q^{-1})_n\,(q^s;q^{-1})_m}{(q^{s+1};q)_{n+m+1}}.$$

Comparing this to (6.7) we see that the polynomial $P_{n,m}$ is, up to a constant factor, the type II multiple little q-Jacobi polynomial with parameters $\alpha_1=\alpha_2=\beta=0$.

9 Irrationality of certain Lambert series

A simple Padé approximation problem used in [5] proves the irrationality of a family of Lambert series:

$$h^{\pm}(q_1,q_2) = \sum_{k=1}^{\infty}\frac{q_1^k}{1\pm q_2^k}$$

where $q_2 = 1/p_2$, $p_2 \in \mathbb{N}\setminus\{1\}$, $q_1 \in \mathbb{Q}$, $0<q_1<1$. The function which has to be approximated is

$$f(z) = \int_0^1 \frac{q_1^{\log_{q_2} x}}{z-x}\frac{d_{q_2}x}{x} = \sum_{k=0}^{\infty}\frac{q_1^k}{z-q_2^k}.$$

Evaluating this function at $z=\pm 1$ gives the desired constants. The approximation problem is to find a polynomial $P_n = \sum_{i=0}^n a_i x^i$ of degree n and a polynomial Q_n of degree $n-1$ such that

$$P_n(z)f(z) - Q_n(z) = \mathcal{O}\left(\frac{1}{z^{n+1}}\right), \qquad z\to\infty.$$

The orthogonality conditions that follow from this problem, are given by

$$\int_0^1 P_n(x)\frac{q_1^{\log_{q_2} x}}{x}x^k d_{q_2}x = 0, \qquad k=0,1,\ldots,n-1.$$

Since the q-Mellin transform of the function $x^j q_1^{\log_{q_2} x}$ is obviously $(1-q_1q_2^{j+s+1})^{-1}$, we get

$$\left(\widehat{\frac{P_n(x)q_1^{\log_{q_2} x}}{x}}\right)(s) = \sum_{i=0}^{n}\frac{a_i}{1-q_1q_2^{i+s}} = \frac{\Pi_n(q_2^s)}{(q_1q_2^s;q_2)_{n+1}}$$

where Π_n is a polynomial of degree n. This polynomial can be determined by using the orthogonality conditions: the q-Mellin transform needs to have zeros at $s=0,1,\ldots,n-1$, hence we conclude that

$$\left(\widehat{\frac{P_n(x)q_1^{\log_{q_2} x}}{x}}\right)(s) = C\frac{\left(q_2^s;q_2^{-1}\right)_n}{(q_1q_2^s;q_2)_{n+1}}.$$

The polynomial P_n that results from this expression, turns out to be a little q-Jacobi polynomial:

$$P_n(z) = {}_2\phi_1\left(\begin{matrix} p_2^n, q_1 q_2^n \\ q_1 \end{matrix}\middle| q_2; q_2 z\right) = \sum_{k=0}^{n} \frac{(p_2^n;q_2)_k (q_1 q_2^n;q_2)_k}{(q_1;q_2)_k (q_2;q_2)_k} q_2^k z^k.$$

References

[1] R. Apéry, *Irrationalité de $\zeta(2)$ et $\zeta(3)$*, Astérisque **61** (1979), 11–13.

[2] A. I. Aptekarev, A. Branquinho, W. Van Assche, *Multiple orthogonal polynomials for classical weights*, Trans. Amer. Math. Soc. **355** (2003), 3887–3914.

[3] K. Ball, T. Rivoal, *Irrationalité d'une infinité de valeurs de la fonction zèta aux entiers impairs*, Invent. Math. **146** (2001), no. 1, 193–207.

[4] F. Beukers, *A note on the irrationality of $\zeta(2)$ and $\zeta(3)$*, Bull. London Math. Soc. **11** (1979), 268–272.

[5] J. Coussement, C. Smet, *Irrationality proof of certain Lambert series using little q-Jacobi polynomials*, submitted.

[6] A. A. Gonchar, E. A. Rakhmanov, *On the equilibrium problem for vector potentials*, Usp. Mat. Nauk **40** no. 4 (244) (1985), 155–156 (Russian); Russ. Math. Surveys **40** no. 4 (1985), 183–184.

[7] A. A. Gonchar, E. A. Rakhmanov, V. N. Sorokin, *Hermite-Padé approximants for systems of Markov-type functions*, Mat. Sb. **188** (1997), 38–58 (Russian); Russian Acad. Sb. Math. **188** (1997), 671–696.

[8] K. Postelmans, W. Van Assche, *Irrationality of $\zeta_q(1)$ and $\zeta_q(2)$*, J. Number Theory **126** (2007), 119-154.

[9] C. Krattenthaler, T. Rivoal, W. Zudilin, *Séries hypergéométriques basiques, q-analogues des valeurs de la fonction zêta et séries d'Eisenstein*, J. Inst. Math. Jussieu **5** (2006), 53-79.

[10] Y. V. Nesterenko, *On the linear independence of functions*, Vestnik Mosk. Univ., Ser. I (1985), 46–54 (in Russian); translated in Mosc. Univ. Math. Bull. **4**0, nr. 1 (1985), 69–74.

[11] E. M. Nikishin, *On the irrationality of the values of the functions $F(x,s)$*, Mat. Sbornik **37** (1979), no. 3, 381–388.

[12] E. M. Nikishin, V. N. Sorokin, *Rational Approximations and Orthogonality*, Translations of Mathematical Monographs **92**, Amer. Math. Soc., Providence RI, 1991.

[13] L. R. Piñeiro, *On simultaneous approximations for a collection of Markov functions*, Vestnik Mosk. Univ., Ser. I (1987), no. 2, 67–70 (in Russian); translated in Moscow Univ. Math. Bull. **42** (1987), no. 2, 52–55.

[14] T. Rivoal, *Irrationalité d'au moins un des neuf nombres* $\zeta(5), \zeta(7), \ldots, \zeta(21)$, Acta Arith. **103** (2002), no. 2, 157–167.

[15] W. Van Assche, *Multiple orthogonal polynomials, irrationality and transcendence*, in "Continued fractions: from analytic number theory to constructive approximation" (Columbia, MO, 1998), Contemporary Mathematics **236** (1999), 325–342.

[16] W. Van Assche; *Analytic number theory and approximation*, Coimbra Lecture Notes on Orthogonal Polynomials (A. Branquinho, A. Foulquié, eds.), Nova Science Publishers, New York, 2008, pp. 197–229.

[17] W. Van Assche, E. Coussement, *Some classical multiple orthogonal polynomials*, J. Comput. Appl. Math. **127** (2001), 317–347.

[18] W. Van Assche, J. S. Geronimo, A. B. J. Kuijlaars, *Riemann-Hilbert problems for multiple orthogonal polynomials*, in 'Special Functions 2000' (J. Bustoz et al., eds.), NATO Science Series II. Mathematics, Physics and Chemistry, Vol. 30, Kluwer, Dordrecht, 2001, pp. 23–59.

[19] V. V. Zudilin, *One of the eight numbers* $\zeta(5)$, $\zeta(7)$,..., $\zeta(17)$, $\zeta(19)$ *is irrational*, Mathematical Notes **70**:3 (2001), 426–431.

[20] V. V. Zudilin, *One of the numbers* $\zeta(5)$, $\zeta(7)$, $\zeta(9)$, $\zeta(11)$ *is irrational*, Uspekhi Mat. Nauk **56** (2001), no. 4 (340), 149–150; Russian Math. Surveys **56** (2001), no. 4, 774–776.

[21] V. V. Zudilin, *Irrationality of values of the Riemann zeta function*, Izv. Ross. Akad. Nauk Ser. Mat. **66** (2002), no. 3, 49–102; Izv. Math. **66** (3) (2002), 489–542.

Walter Van Assche (corresponding author)
Christophe Smet
Department of Mathematics
Katholieke Universiteit Leuven
Celestijnenlaan 200 B - box 2400
BE-3001 Leuven
BELGIUM
walter@wis.kuleuven.be
christophe@wis.kuleuven.be